



SOULMAP:

A Vietnamese Astrology Question Answering System using RAG

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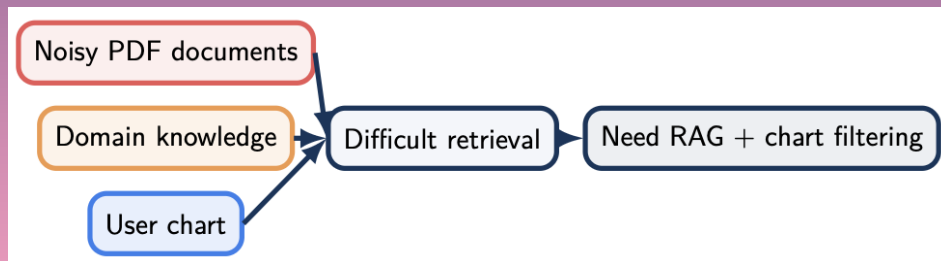
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Motivation

Main message: General LLMs may hallucinate in specialized domains.

- Specialized QA requires **trusted knowledge sources**.
- Vietnamese astrology documents are long, semi-structured, and noisy.
- Answers should depend on documents **and** the user's personalized chart.



Problem Statement

Task: Build a Vietnamese QA system that uses both a personalized chart and retrieved evidence.

Component	Description
Input	Name, gender, birth date, birth time, question
Intermediate output	Personalized astrology chart
Knowledge source	Processed Vietnamese astrology PDFs
Final output	Grounded Vietnamese answer

Core idea:

Instead of asking the LLM directly, SoulMap generates the user's chart, retrieves relevant knowledge, and constructs a grounded prompt for Gemini.

Related Work / Background

Main message: SoulMap combines domain-specific QA, RAG, and Vietnamese dense retrieval.

Topic	Description
Domain-specific QA	Answers must be grounded in a predefined knowledge source
Retrieval-Augmented Generation	Retrieves documents before answer generation
Vietnamese embedding model	Encodes Vietnamese queries and documents for semantic retrieval

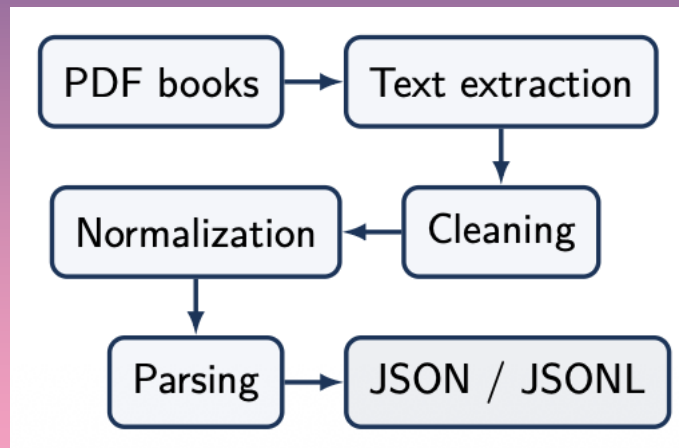
- RAG helps reduce unsupported answers.
- The knowledge base can be updated without retraining the LLM.
- Dense retrieval supports Vietnamese semantic matching.

Dataset and Knowledge Sources

Main message: The knowledge base is built from Vietnamese astrology books in PDF format.

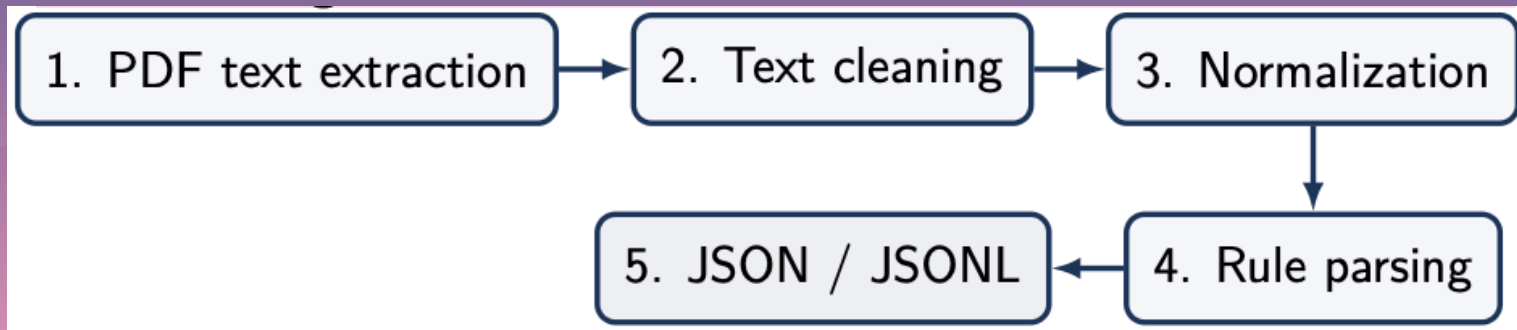
- *Tu Vi Thuc Hanh*
- *Tu Vi Dau So Tan Bien*
- *Tu Vi Tong Hop*

Documents cover palaces, stems-branches, destiny elements, star placement, brightness, major periods, and interpretation rules.



Preprocessing Pipeline

Main message: PDFs must be converted into clean, retrievable data.



Why normalization matters?

The same palace or star may appear in multiple text formats. Normalized identifiers include `cung_menh`, `cung_phu_mau`, and `cung_thien_di`.

Chunking Strategies

Main message: Two chunking strategies balance recall and precision.

Strategy	Goal	Strength	Limitation
Fixed-size chunks	Preserve broad textual context	High recall; easy to construct	May break logical boundaries
Structured chunks	Represent astrology rules	High precision; supports chart filtering	Requires reliable parsing

Fixed-size chunks

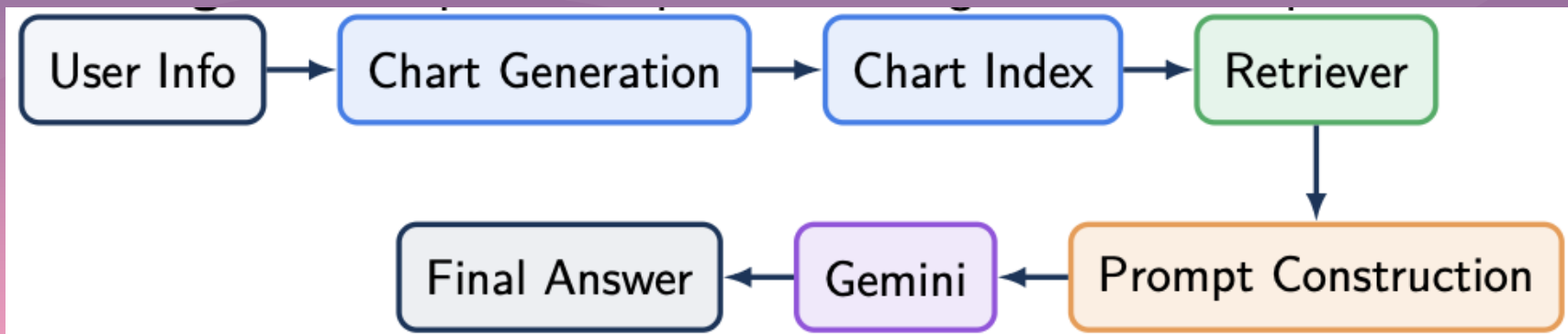
Broad context and higher recall

Structured chunks

Palace-star rules and chart-specific precision

System Overview

Main message: SoulMap has two phases: chart generation and question answering.



- Chart initialization creates personalized context.
- QA uses chart index, retrieved chunks, grounded prompt, and Gemini.

User Chart Generation

Main message: The personalized chart is central context for retrieval and generation.

- Convert solar date to lunar date.
- Compute heavenly stem, earthly branch, destiny element, and configuration.
- Determine the twelve palaces.
- Store stars, brightness levels, lifecycle stage, Tuần-Triệt markers, and major periods.

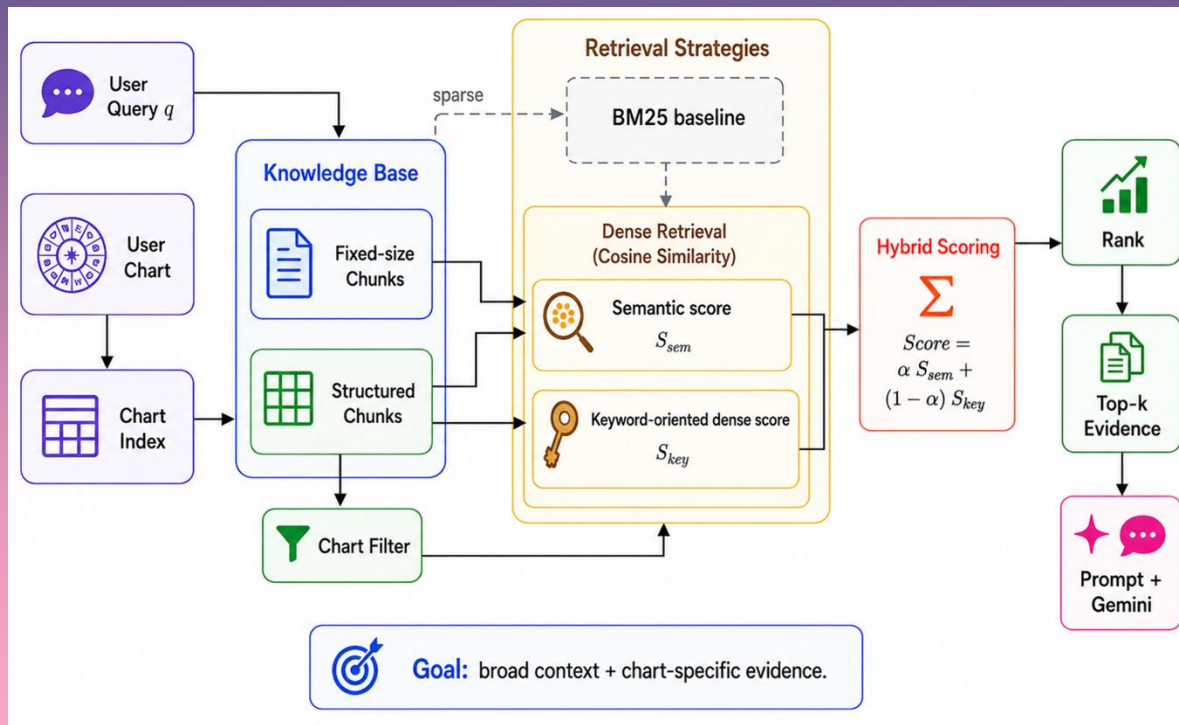
Twelve houses

Mệnh, Phụ Mẫu, Phúc Đức, Điền Trạch, Quan Lộc, Nô Bộc, Thiên Di, Tật Ách, Tài Bạch, Tử Tức, Phu Thê, Huỳnh Đệ

Retriever Design

Main message: Retrieval must consider the question and the personalized chart.

- **Input:** user query + chart index.
- **Candidates:** fixed-size chunks + structured chunks.
- **Filtering:** palace/star compatibility.
- **Ranking:** semantic score + keyword-oriented dense score.



Retrieval Modes

Main message: Five modes compare retrieval strategies.

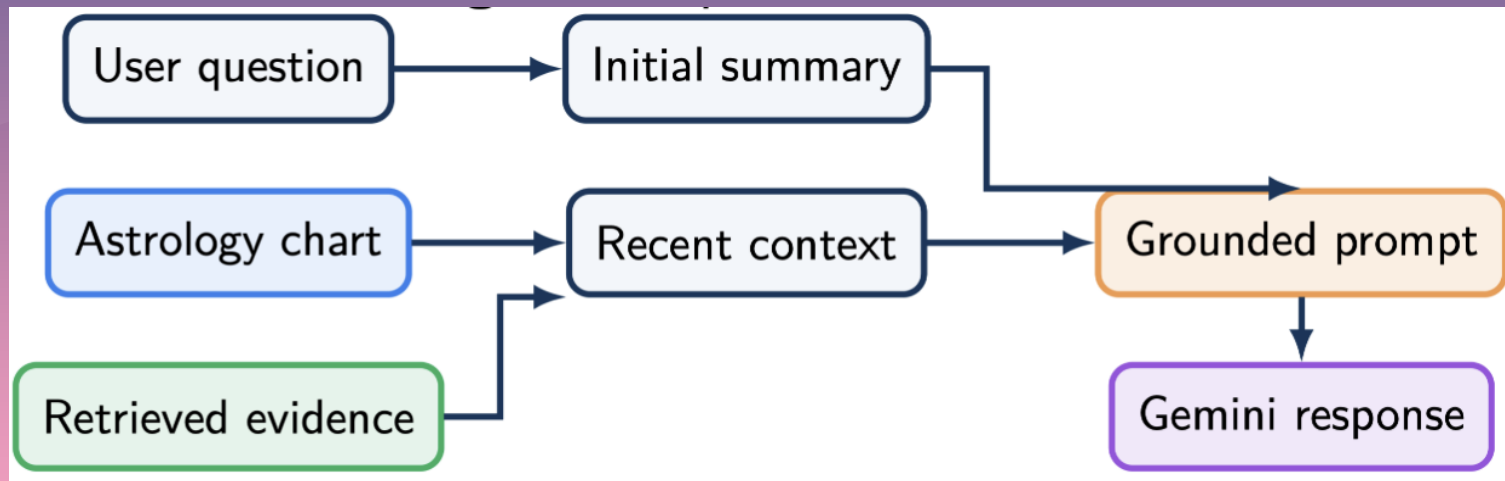
Mode	Data Source	Purpose
fixed_similarity	Fixed_size_chunk	Dense retrieval baseline
fixed_structured_similarity	Fixed + structured chunks	Combines context and rules
fixed_structured_keyword	Fixed + structured chunks	Default hybrid mode
structured_similarity	Structured chunks only	Chart-filtered semantic retrieval
structured_keyword	Structured chunks only	Metadata-focused retrieval

Default Intuition

fixed_structured_keyword combines recall from fixed-size chunks and precision from structured metadata.

Prompt Construction

Main message: Prompt construction connects chart, evidence, and Gemini.



safe-framing: reference-based interpretations grounded in retrieved context, not deterministic prediction.

Experimental Setup

Main message: Experiments evaluate retrieval and answer generation quality.

Component	Configuration
Task	Domain-specific QA
Domain	Vietnamese astrology
Method	RAG (Lewis et al., 2020)
Backbone LLM	Gemini-2.5-flash
Retriever encoder	BKAI Vietnamese bi-encoder
Retriever type	Hybrid semantic-keyword
Retrieval data	Structured + fixed chunks
Top- k	8
α	0.75
Input context	User chart + retrieved chunks
Output language	Vietnamese

Evaluation Dataset and Metrics

Main message: Evaluation includes retrieval-level and generation-level metrics.

Retrieval evaluation

Feature Recall, Query Success, Hit@K, MRR

Generation evaluation

Perplexity, Coverage, Strict Grounding, Prompt Grounding

- Synthetic evaluation dataset: **100 users and 108 questions**.
- Questions are grouped into easy, medium, and hard levels.
- Difficulty depends on the number of relevant chart features.

Retrieval Results

Main message: Structured similarity ranks best, while hybrid retrieval improves coverage.

Method	Recall	Success	MRR	Hit@1	Hit@3
Fixed Similarity	0.525	0.352	0.472	0.380	0.500
Fixed + Struct Sim	0.548	0.365	0.483	0.384	0.514
Fixed + Struct Keyword	0.546	0.363	0.483	0.383	0.515
Structured Similarity	0.522	0.354	0.528	0.436	0.593
Structured Keyword	0.416	0.270	0.421	0.352	0.452

Table 6: Retrieval performance across different retrieval configurations.

Interpretation

There is no single best retrieval mode across all metrics: structured similarity improves ranking, while fixed + structured similarity improves coverage and robustness.

Retrieval Results

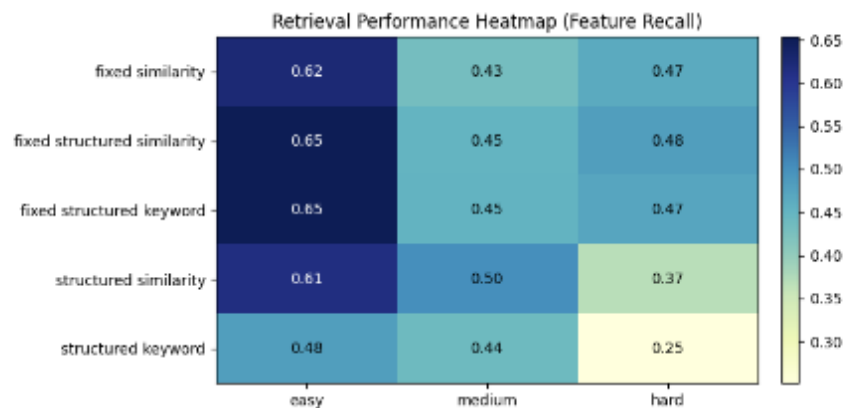


Figure 3: Feature Recall Retrieval Performance Across Modes and Difficulty Levels

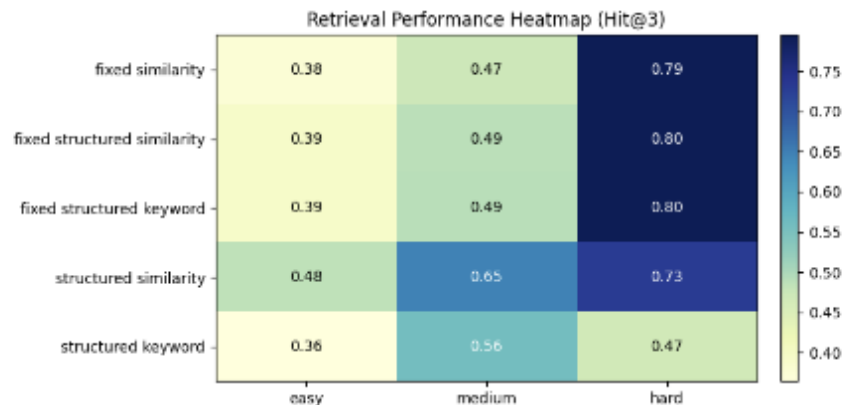


Figure 4: Hit@3 Retrieval Performance Across Modes and Difficulty Levels

Generation Results

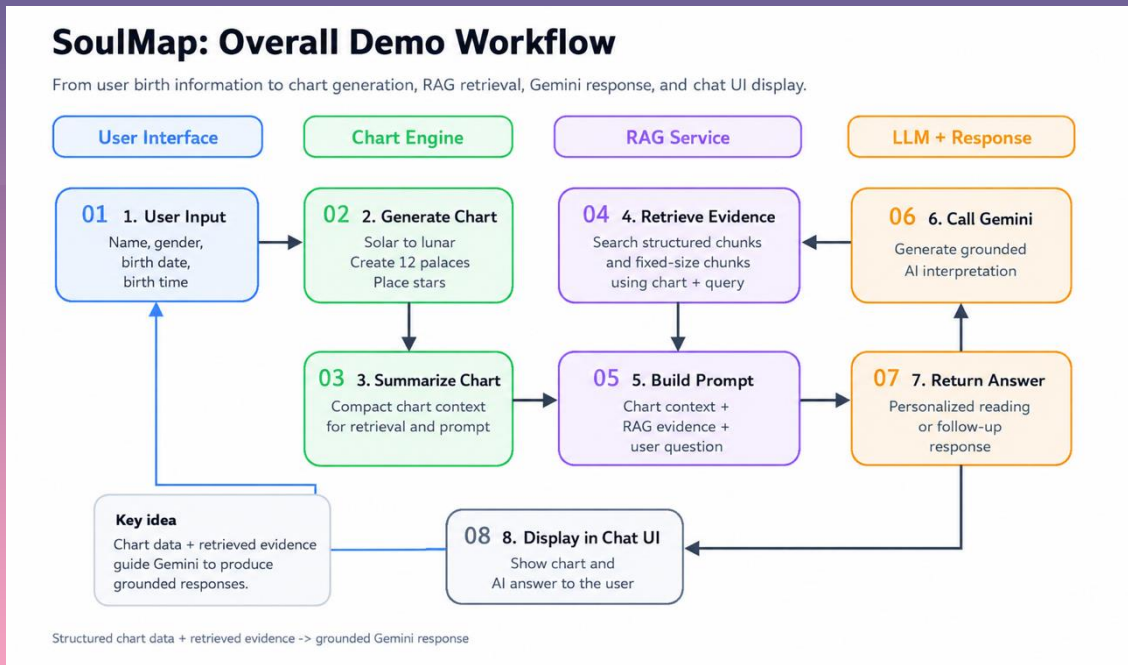
Main message: Fixed structured similarity strongly improves coverage.

Metric	Structured Similarity	Fixed Structured Similarity	Delta
PPL	17.48	17.70	+0.23
Coverage	0.475	0.829	+0.354
Strict Grounding	0.400	0.422	+0.022
Prompt Grounding	0.996	0.998	+0.002

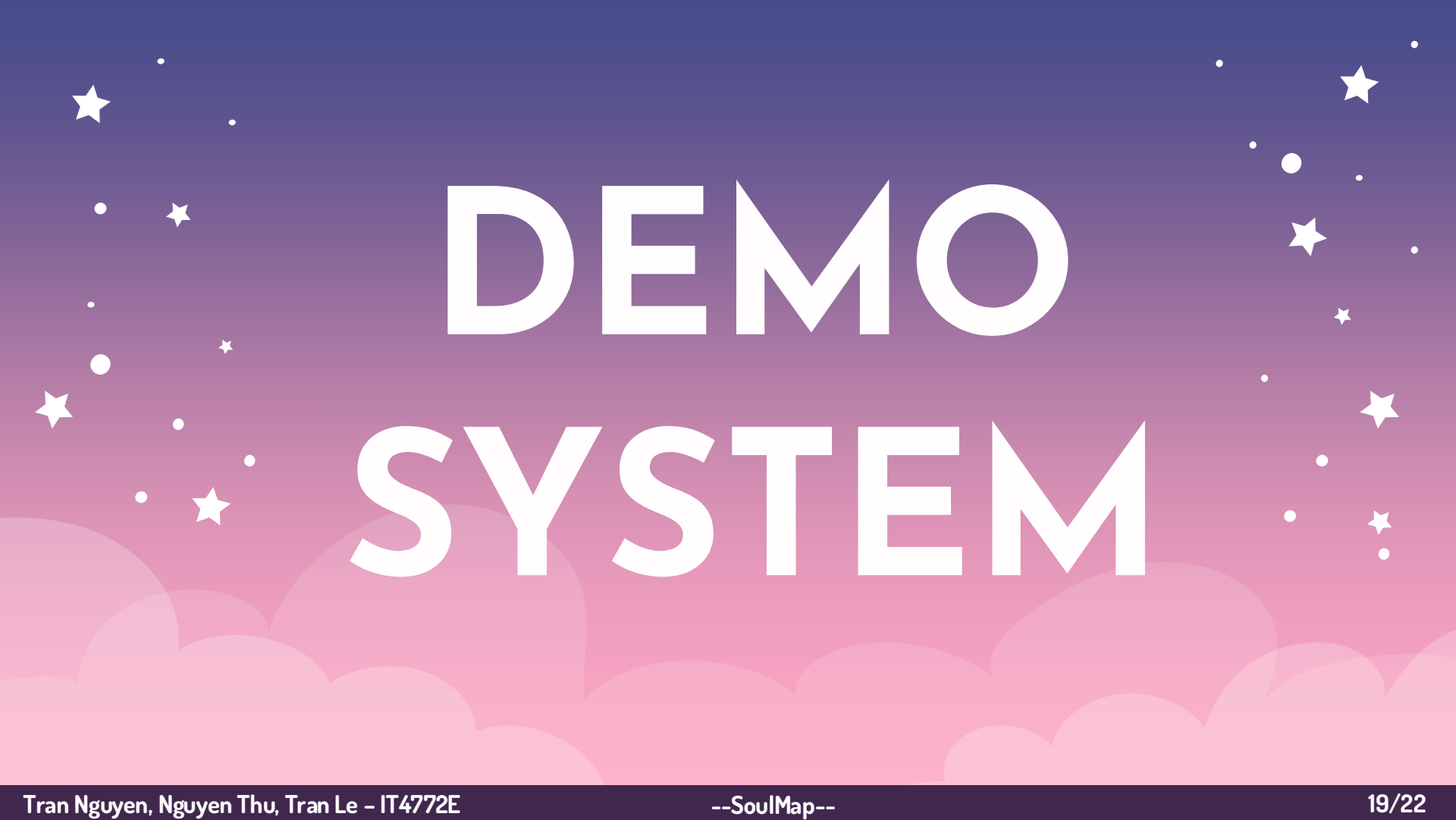
Table 7: Comparison of structured similarity and fixed structured similarity across evaluation metrics.

Demo Workflow

Main message: The demo runs an end-to-end pipeline before Gemini answers.



The LLM does not answer directly: each question goes through chart generation, retrieval, and prompt construction first.



DEMO SYSTEM

Limitations

Main message: The system depends on document quality, parsing quality, benchmark availability, and the LLM.

- PDF extraction noise.
- OCR and formatting errors.
- Regex parser coverage limits.
- Lack of a large labeled benchmark.
- Dependency on Gemini.
- Latency and cost at scale.
- Astrology is interpretive.
- Avoid absolute predictive claims.

Conclusion and Future Work

Main message: RAG supports personalized, document-grounded Vietnamese astrology QA.

Conclusion

- Combines chart generation, PDF knowledge construction, and hybrid retrieval.
- Structured + fixed-size chunking balances precision and recall.
- Initial results show strong coverage gains with RAG.

Future Works

- Clean structured dataset and build a gold benchmark.
- Add a reranker and evidence citation.
- Optimize embedding and retrieval caches.
- Complete the final demo interface.



THANKS!

DO YOU HAVE ANY QUESTIONS?

Takeaway

SoulMap uses RAG, Vietnamese dense retrieval, and chart-based filtering to provide reference-grounded astrology interpretations.